

Programming Using C++ Level 1

Arrays



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Arrays

Definition

- ◆ An array is a collection of multiple values of the same data type stored under one variable name.
- ◆ Each value in an array is accessed using an index.
- ◆ Array indexes start from 0.

Explanation

- ◆ This program creates an array called *Grades* that can store 3 float values.
- ◆ The elements are accessed using indexes:
 - ▣ *Grades*[0] → first grade
 - ▣ *Grades*[1] → second grade
 - ▣ *Grades*[2] → third grade
- ◆ The program reads the three grades and calculates their average.

Steps in main()

1. Declare the array

◆ Create an array that can store 3 float values.

2. Enter the first grade

◆ Store the value in `Grades[0]`.

3. Enter the second grade

◆ Store the value in `Grades[1]`.

4. Enter the third grade

◆ Store the value in `Grades[2]`.

5. Calculate the average

◆ Add the three array elements and divide the result by 3.

6. Display the result

◆ Print the calculated average using `cout`.

💡 Example

◆ Array declaration:

```
float Grades[3];
```

◆ The array contains 3 elements:

```
Grades[0]
```

```
Grades[1]
```

```
Grades[2]
```

◆ If the grades are:

```
Grades[0] = 80
```

```
Grades[1] = 90
```

```
Grades[2] = 70
```

◆ The average is 80.

➡ Result

◆ If the user enters:

```
Grade1 = 80
```

```
Grade2 = 90
```

```
Grade3 = 70
```

◆ The program displays:

```
The average of grade is: 80
```

Note

- ◆ The number inside the brackets when declaring an array represents its size.
- ◆ `float Grades[3]` creates space for 3 float elements.
- ◆ The first index is always 0.
- ◆ For an array of size 3, the valid indexes are 0, 1, and 2.

⊘ Warning

- ◆ Do not access an index outside the array's valid range.
- ◆ For `Grades[3]`, `Grades[3]` is invalid because the last valid index is `Grades[2]`.



Comparison



Array

- ✓ Stores multiple values of the same data type.
- ✓ Uses one variable name with different indexes.
- ✓ Makes it easier to work with collections of related values.



Separate Variables

- ✓ Each value needs its own variable.
- ✓ Values have different variable names.
- ✓ Managing many related values becomes harder.

✦✦ Summary

- ✓ An array stores multiple values of the same data type.
- ✓ Array indexes start from 0.
- ✓ Arrays make it easier to store and work with collections of related values.

Thank you 😊